Project Documentation

Project Title: Walmart Retail Sales Analysis using Python, Pandas, Matplotlib, Seaborn, and Plotly

1. Project Overview

This project focuses on analyzing Walmart retail sales data using Python. The analysis includes data cleaning, preprocessing, exploratory data analysis (EDA), and data visualization to identify sales patterns, profitability trends, customer behaviour, and regional performance. The goal is to derive meaningful business insights that can support data-driven decision-making.

2. Tools Used

- Python – Data analysis and processing
- Pandas – Data cleaning and manipulation
- NumPy – Numerical computations
- Matplotlib – Data visualization
- Seaborn – Statistical visualizations
- Plotly – Interactive visualizations
- Jupyter Notebook – Development environment

3. Dataset

Source: [Walmart Retail Dataset]

Dataset Size:

- Rows: 1,037,247
- Columns: 23

Key Features:

- Order Date
- Customer ID
- Product Category
- Sales
- Profit
- Quantity
- Discount
- Unit Price
- Shipping Cost
- Region

- State
- City
- Shipping Mode

4. Steps Followed

1. Data Loading

- Imported the dataset into Jupyter Notebook using Pandas.
- Verified dataset dimensions and structure.

2. Data Cleaning and Preprocessing

- Converted date columns into datetime format.
- Filtered records from 2021 onwards.
- Handled missing values in numerical and categorical columns.
- Removed duplicate records.
- Standardized text values.
- Performed final validation checks.

3. Exploratory Data Analysis (EDA)

- Analyzed sales and profit distributions.
- Examined product category performance.
- Studied customer segment behaviour.
- Evaluated regional sales performance.
- Conducted correlation analysis among numerical variables.

4. Data Visualization

Created visualizations including:

- Histograms
- Boxplots
- Bar Charts
- Pie Charts
- Scatter Plots
- Correlation Heatmaps
- Interactive Plotly Charts

5. Insight Generation

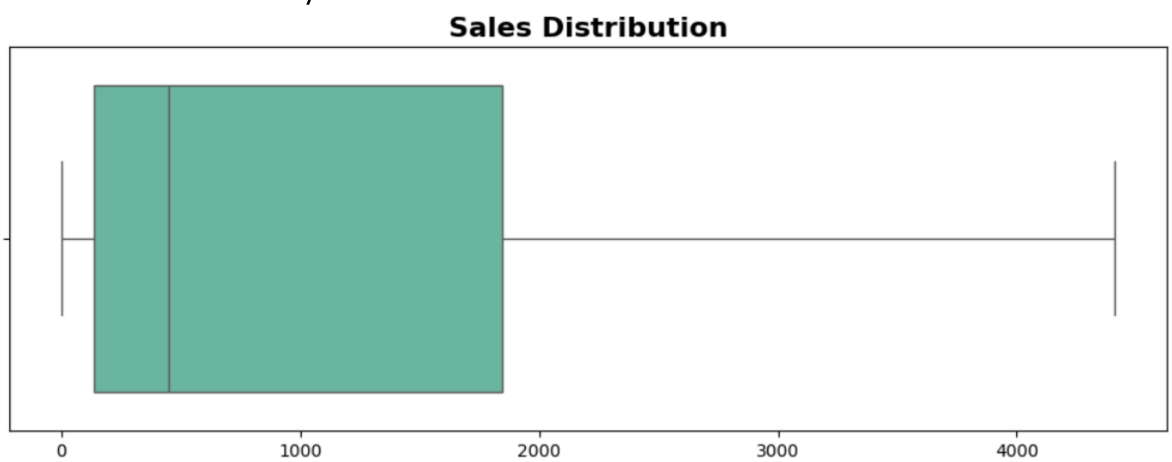
Interpreted patterns and trends to derive business recommendations.

5. Key Insights

• Most sales transactions are concentrated within lower sales ranges.
• Office Supplies recorded the highest transaction volume among product categories.
• East and Central regions generated the highest sales volumes, indicating stronger market demand compared to other regions.
• Profit values showed substantial variation, with both highly profitable and loss-making transactions present in the dataset.
• Sales and profit showed a positive relationship, although high sales did not always result in high profits.
• Discounts exhibited only a weak correlation with profit, indicating additional factors influence profitability.

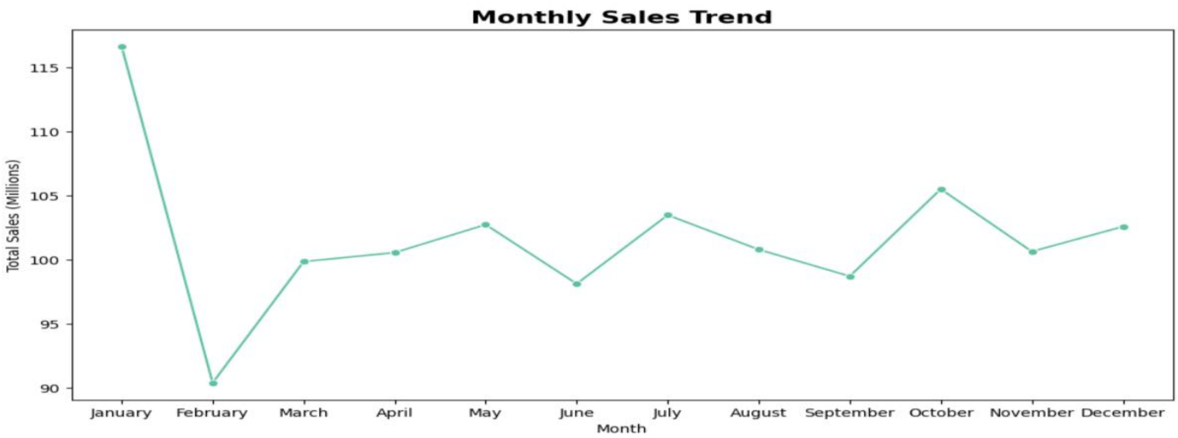
6. Visualizations

• Sales Distribution Analysis



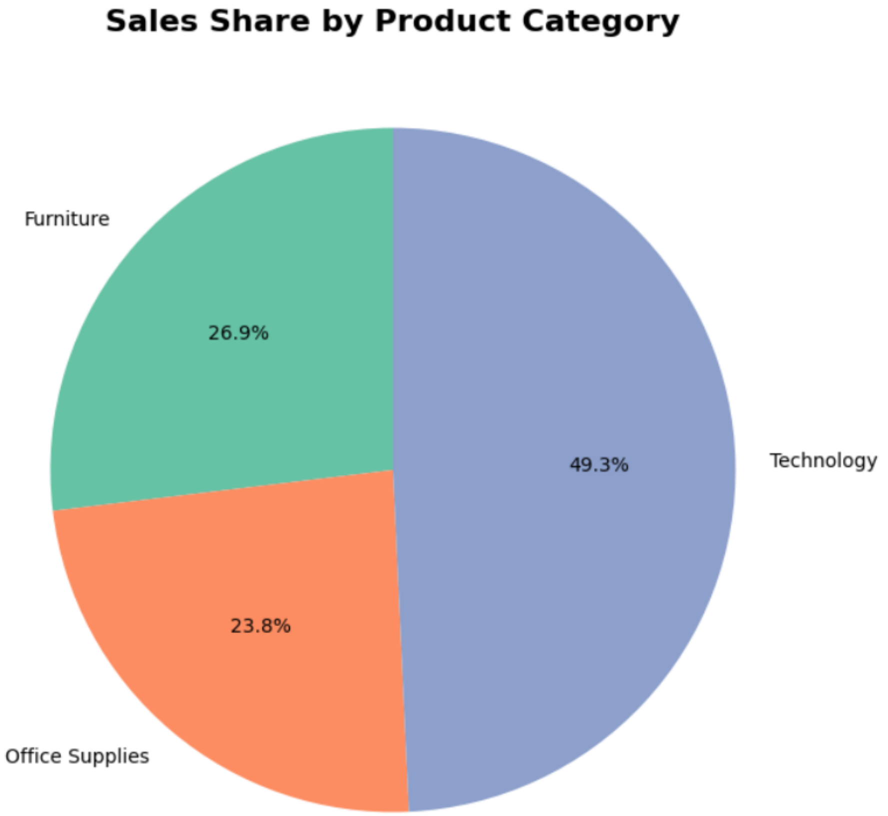
Insight: Most sales transactions fall within lower sales ranges, while a small number of high-value orders contribute significantly to overall revenue.

• Profit Distribution Analysis



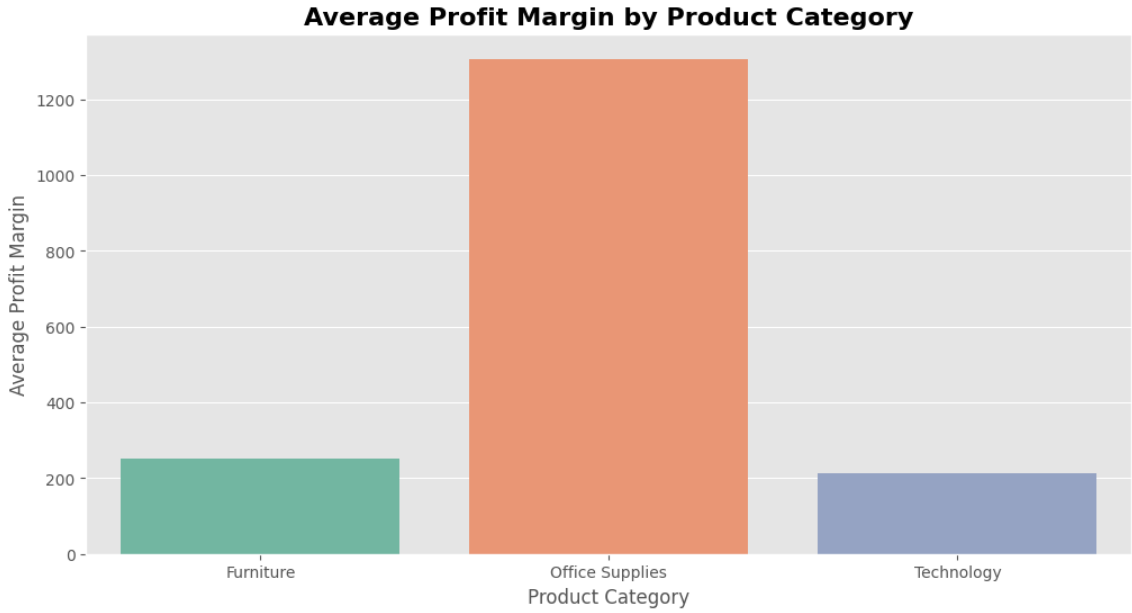
Insight: Profit values show substantial variation across transactions, with both highly profitable and loss-making orders present in the dataset.

• Product Category Distribution



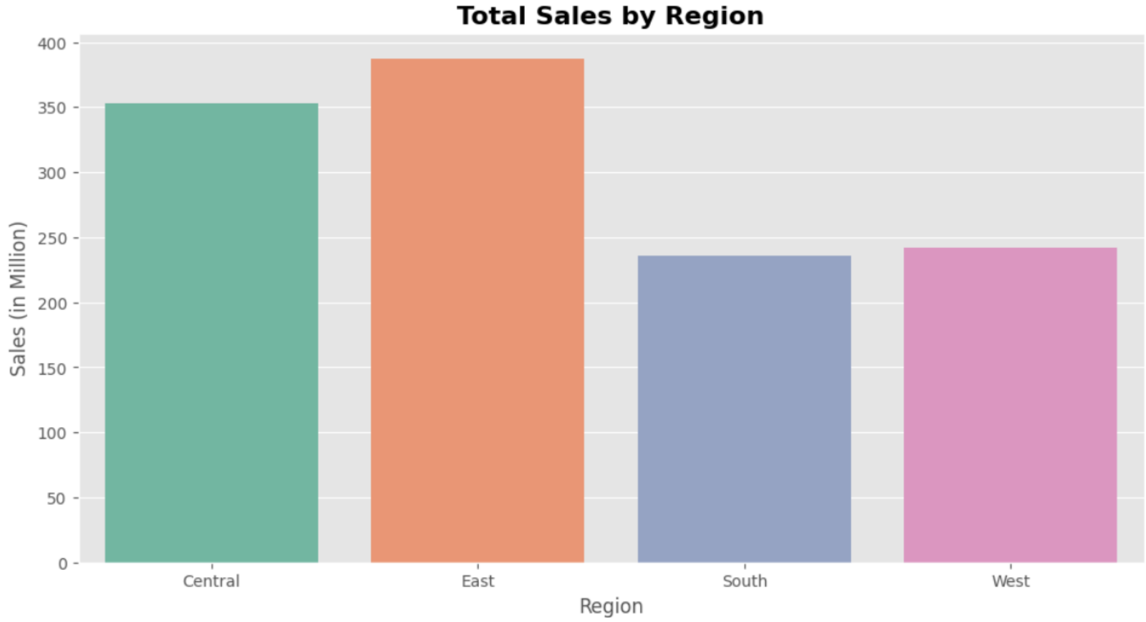
Insight: Office Supplies contributed the largest share of total sales, followed by Technology and Furniture, highlighting its dominant role in overall revenue generation.

• Average Profit Margin by Product Category



Insight: Profit margins vary across product categories, indicating differences in pricing strategies, operational costs, and overall profitability. Categories with higher margins may present stronger opportunities for business growth.

• Regional Sales Comparison



Insight: The East and Central regions generated the highest sales volumes, indicating stronger market performance compared to other regions.

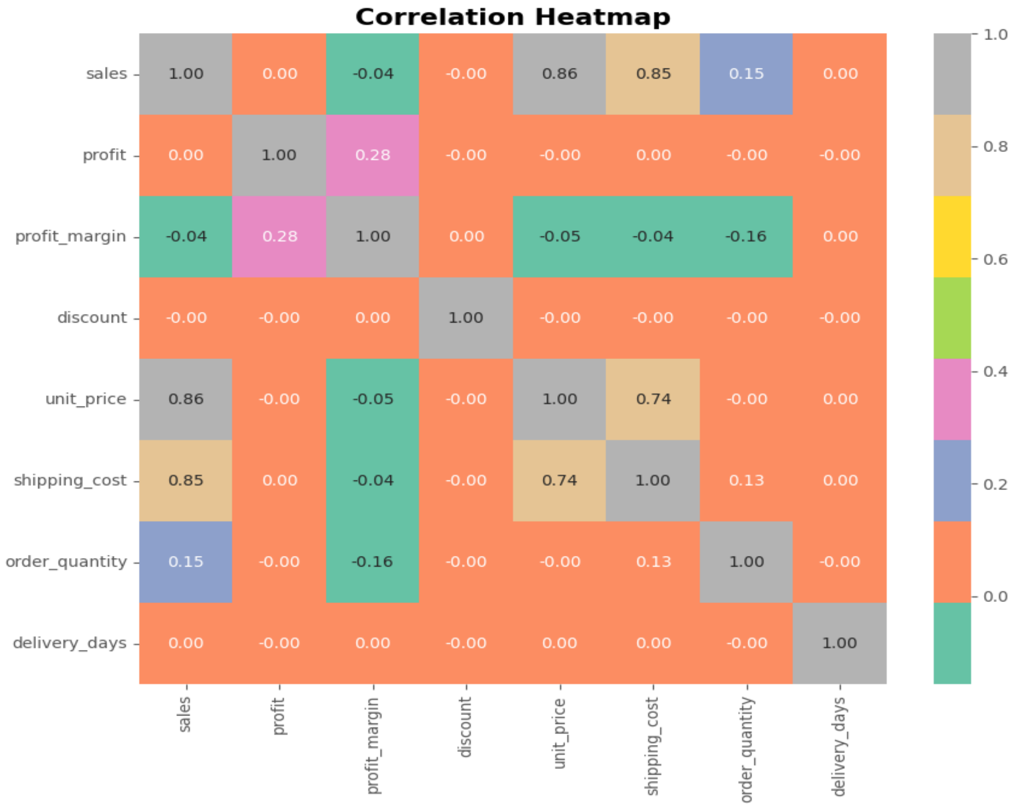
• Product Category Distribution Across Regions

Insight: Office Supplies consistently recorded the highest transaction volume across regions, while Furniture and Technology displayed varying regional demand patterns.

• Discount vs Profit Relationship

Insight: Discounts exhibit only a weak relationship with profit, suggesting that profitability is influenced by multiple factors beyond discount levels alone.

• Correlation Heatmap



Insight: Sales, unit price, and shipping cost show positive relationships, while discount demonstrates minimal correlation with profit.

7. Files Included

• Walmart-Retail-Dataset.csv – Dataset
• Walmart-Retail-Analysis.ipynb – Complete analysis notebook
• README.md – Project overview and instructions
• Walmart-Retail-Analysis-Documentation.pdf – Project summary document
• Walmart_logo.png – Logo used in project

8. How to Use

1. Open the Jupyter Notebook file.
2. Run the cells sequentially to perform data cleaning, analysis, and visualization.
3. Review the generated charts and insights.
4. Modify the analysis to explore additional business questions if required.

9. Conclusion

This project demonstrates an end-to-end data analytics workflow using Python. Through data cleaning, exploratory analysis, and visualization, valuable insights were obtained regarding sales performance, profitability, customer behavior, and regional trends. The project highlights how data analytics can support informed business decisions and serve as a foundation for future predictive analytics initiatives.